

CY 510-01 OrbitSec: STRIDE Threat Model

<https://github.com/ctarique/orbitsec>

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Introduction

To map our engineering work to formal security theory, this document provides a comprehensive threat model of the OrbitSec architecture. The system utilizes a "Verify-then-Decrypt" paradigm to protect against the following STRIDE threat categories.

STRIDE Category	Threat Vector	OrbitSec Mitigation Strategy
Spoofing	An unauthorized actor attempts to masquerade as the legitimate transmission source.	ECDSA Public-Key Verification: Validates the signature before initializing the decryption engine.
Tampering	An attacker intercepts the transmission and alters the ciphertext or IV in transit.	SHA-256 Cryptographic Hash: Any bit-level modification alters the hash, causing verification to fail.
Repudiation	The transmitting entity denies having sent a specific telemetry payload or command.	Asymmetric Private Key: Only the true possessor of the private key could have generated the valid packet.

STRIDE Category	Threat Vector	OrbitSec Mitigation Strategy
Information Disclosure	An unauthorized observer eavesdrops on the communication channel.	AES-256-CBC Encryption: Secures the plaintext using a 256-bit symmetric key.
Denial of Service	An attacker floods the receiving device with malformed data to exhaust its memory.	Efficient Memory Management: Rejects invalid signatures immediately and frees memory without passing data to the AES engine.
Elevation of Privilege	An attacker crafts an oversized payload designed to overflow the C firmware's memory buffer.	Strict Bounds Checking: Enforces rigid limits on payload and signature lengths before copying any data.